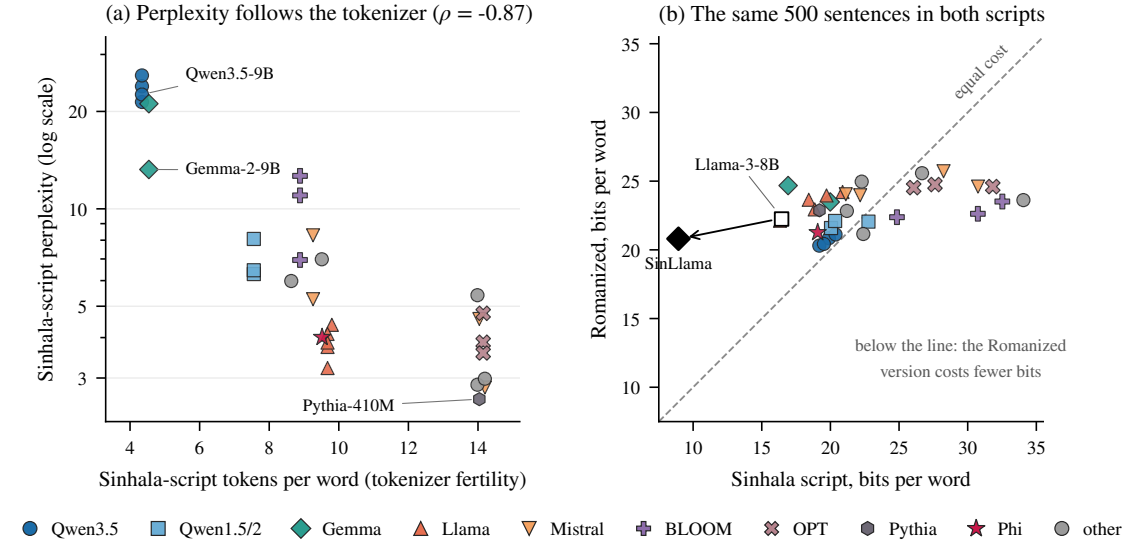
(a) Perplexity follows the tokenizer (ρ = -0.87)
Sinhala-script perplexity (log scale)
Sinhala-script tokens per word (tokenizer fertility)
Qwen3.5-9B
Gemma-2-9B
Pythia-410M
20
10
5
3
4
6
8
10
12
14
(b) The same 500 sentences in both scripts
Romanized, bits per word
Sinhala script, bits per word
equal cost
Llama-3-8B
SinLlama
below the line: the Romanized
version costs fewer bits
35
30
25
20
15
10
Qwen3.5
Qwen1.5/2
Gemma
Llama
Mistral
BLOOM
OPT
Pythia
Phi
other